



ST3131 Midterm Paper 2026

Regression Analysis (National University of Singapore)



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NATIONAL UNIVERSITY OF SINGAPORE

Department of Statistics and Data Science

ST3131 - Regression Analysis

(Semester 2 AY2025/2026)

Midterm Test

Time Allowed: 80 minutes

INSTRUCTIONS TO STUDENTS

1. Students are required to complete this test individually.
2. This is an OPEN BOOK BLOCK INTERNET exam. You may refer to your lecture notes, tutorials, textbooks or any notes that you have made, but you are not allowed to perform any online search.
3. The use of offline Large Language Models (LLMs) is prohibited for this test.
4. Both programmable calculators and non-programmable calculators are allowed.
5. Students are required to answer ALL questions. Total mark is 50.
6. **Hand-in your answer booklet to the invigilators at the end of the test.**
You do not need to submit the code file.
7. Report numerical answers to at least three significant figures if it's absolute is smaller than one (e.g. 0.0123) and to three decimal places if it's absolute is larger than one (e.g. -2.345).

Recall some definitions:

For a sample $(x_1, \dots, x_n)$, sample variance is $\frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})^2$.

For two samples $(x_1, \dots, x_n)$ and $(y_1, \dots, y_n)$, the sample covariance is

$$Cov(x, y) = \frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})$$

and the correlation coefficient is

$$Corr(x, y) = \frac{Cov(x, y)}{\sqrt{Var(x)}\sqrt{Var(y)}}.$$

1. (*40 points*) The data file `volkswg-1k.csv` contains data from a random sample of 1000 used Volkswagen car listings in a country that were sold in 2021. The cars in the data set were registered from 2004 to 2020.

The given file contains columns with their names listed below.

Column's Name	Description
year	year the car was registered (getting number plate)
price	price of the car listing (in £)
transmission	transmission type (Automatic/Semi-Auto/Manual)
mileage	miles travelled
fuelType	type of fuel used (Petrol/Diesel/Hybrid/Other)
tax	road tax of the car (in £)
mpg	miles per gallon
engineSize	size of engine (in litres)

In all the tests below, please use significance level $\alpha = 0.05$.

Load the file `volkswg-1k.csv` into R and name it as **data**. Our purpose is to fit a linear model that helps to predict car's price using the 7 input variables.

- (2 points) Using the data frame **data**, add a new column (called **age**) that gives each car's age at the time it was sold in 2021 (i.e., 2021 minus the registration year).
Report the number of cars that has age larger than 10 when it was sold.
- (3 points) For each type of transmission, compute the sample correlation between **price** and **age**.
Report the correlation values.
- (2 points) Create a histogram for **price** and comment if it is suitable to be the response for a linear regression model. Explain.
- (2 points) Plot a scatter plot between **price** and **mileage**. Compute and report the correlation coefficient between them. Give your comments.
- (3 points) Fit a multiple linear model for **price** that depends on all the 7 input variables **age**, **transmission**, **mileage**, **fuelType**, **tax**, **mpg** and **engineSize**, name it as **M1**.
Write down the fitted equation for model M1.
- (2 points) Test the significance of variable **tax** in model **M1** by reporting: the hypotheses, the test statistic value, the null distribution and the p-value.
- (4 points) Test the significance of variable **transmission** in model **M1** by reporting: the hypotheses, the test statistic value, the null distribution and the p-value.
- (2 points) Use the fitted equation of model M1, calculate the difference in the price of a Hybrid car and a Petrol car if all other information of them is the same. Report the difference.

9. (4 points) Let **SR1** denotes the standardized residuals of model **M1**. Give your comments on the adequacy of model **M1** based on the residual plots.
10. (2 points) Using the boxplot of **SR1**, does model **M1** have any outliers? If yes, report the number of outliers.
Any data point that has its Cook's distance larger than 1 is considered as an influential point of the model. With that, does model **M1** have any influential points? If yes, report the index of influential points in the data set given.
11. (2 points) Consider a new linear regression model, called **M2**, which has the same set of input variables as model **M1** but the response is the square root of **price**.
Fit model **M2** and report the adjusted R^2 of **M2**.
12. (12 points) Given the two cars, A and B, listed below. Before using model **M2** to predict the selling price for each of them, one would need to check if any of them is extrapolation.
A: year = 2020; transmission = Automatic; mileage = 3000; fuelType = Petrol; tax = 150; mpg = 50; engineSize = 2.
B: year = 2015; transmission = Semi-Auto; mileage = 30000; fuelType = Diesel; tax = 180; mpg = 60; engineSize = 2.
(a) Using only the quantitative input variables of model **M2**, form Hat matrix for the model, denoted as **H**. Report the value of the largest diagonal h_{ii} of **H**, denote it as **h.max**.
(b) With the two new points given above, compute and report the value h_{00} for each point. Is any of the two cars given an extrapolation?
Hint: You could refer to Slide 50 – 54 of Topic 2 - Description.
(c) For car B, using model **M2**, report the predicted selling price and a 90% confidence interval for its selling price.

2. (10 points) A data set is collected with n observations, $(x_1, y_1), \dots, (x_n, y_n)$. A simple linear model $y = \beta_0 + \beta_1 x + \epsilon$ is fitted using the data set above by OLS method. The equation of the fitted model is $\hat{y} = \hat{\beta}_0 + \hat{\beta}_1 x$, where $\hat{\beta}_0$ and $\hat{\beta}_1$ are the estimators of β_0 and β_1 by OLS method.

- (a) (5 points) Show that

$$\hat{\beta}_1 = \frac{Cov(x, y)}{Var(x)}.$$

- (b) (5 points) Given $Var(\hat{y}) = \hat{\beta}_1^2 Var(x)$. Show that $Cor(y, \hat{y}) = |Cor(y, x)|$.

Note: Given a fitted model with $\hat{y} = \hat{\beta}_0 + \hat{\beta}_1 x$, then the covariance

$$Cov(y, \hat{y}) = Cov(y, \hat{\beta}_0 + \hat{\beta}_1 x)$$

is taken with respect to x and y where $\hat{\beta}_0$ and $\hat{\beta}_1$ are constants.

END OF QUESTIONS